Tribhuvan University
Faculty of Humanities and Social Science

INVENTORY MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted to
Department of Computer Application
Milton International College

In partial fulfilment of the required for the Bachelors in Computer Application

Submitted by:
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SUPERVISOR'S RECOMMENDATION

I hereby recommend that this project prepared under my supervision by **Nikesh Manandhar** and **Sudip Magar** entitled "**INVENTORY MANAGEMENT SYSTEM**" in partial fulfilment of the requirements for the degree of Bachelors of Computer Application is recommended for the final evaluation.

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LETTER OF APPROVAL

This is to certify that this project is prepared by **Nikesh Manandhar** and **Sudip Magar** entitled "**INVENTORY MANAGEMENT SYSTEM**" in partial fulfilment of the requirements for the degree of Bachelors of Computer Application has been evaluated. In our opinion it is satisfactory in the scope and quality as a project for the required degree.

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ACKNOWLEDGEMENT

Our earnest appreciation to our Supervisor **Zatil Shrestha** for directing us all through the arranging and advancement period of the framework. Without his vital rule and direction, we would not have achieved the last phase of the improvement.

We would like to thank our close peers and classmates for being supportive and encouraging throughout our project development journey here in Milton International College. Our sincere gratitude goes to Your College Name for providing the platform where we were able to cherish and nurture our compassion and improve our technical skills. Much appreciation to Milton International College for great support and all necessary academic materials that has taught us great helpful academic as well as life lessons.

Yours Sincerely
Nikesh Manandhar
Sudip Magar

ABSTRACT

Designed to enable companies to effectively manage the storage, tracking, and flow of products, an Inventory Management System (IMS) is a web-based tool. It helps store owners or managers to update inventory records, track stock levels, and consistently manage sales and purchases.

Hand-managing inventory can be quite laborious and awkward. It usually entails difficult chores like monitoring products, correcting volumes, maintaining stock in/out records, and producing reports. Data loss, human mistake, missing documents, and retrieval latency make manual systems prone to data loss, so rendering the procedure ungainly and ineffective.

A digital solution for these issues is provided by the Inventory Management System. Real-time inventory tracking is made possible, updates are automated, and accurate and fast access to vital information is made available. It enables store managers to effectively manage inventory, create reports, and keep stock levels. Reliable, user-friendly, simple, the system drastically lowers manual labor, therefore saving time and energy while increasing accuracy and productivity.

Keywords: *Inventory Management System (IMS), Real-time Tracking, Laravel Framework, Database Management, Automation, Stock Control, Web-based Application*

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CHAPTER 1: INTRODUCTION

1.1 Introduction

The Inventory Management System (IMS) is a robust and modern web application developed to support businesses in streamlining their stock control, optimizing warehouse operations, and maintaining accurate real-time inventory data. Designed as a centralized command center, the IMS empowers business owners and inventory managers to make proactive decisions by tracking stock levels, monitoring product movement, and analyzing sales trends. This helps minimize manual stock-taking errors and reduces the risks associated with overstocking or stockouts in traditional inventory management. One of the key goals of the IMS is to bridge the gap between disjointed spreadsheets and a unified, automated system by providing a centralized platform that caters to a wide range of inventory needs.

Businesses can access detailed product catalogs that offer comprehensive information on items, including categories, suppliers, cost prices, and selling prices. Stock transaction history is also meticulously tracked, enabling users to monitor all stock inflows and outflows, view audit trails, and generate insightful reports for better decision-making. These reports are organized into specific types, such as low-stock alerts and sales summaries, making it easier for users to analyze business performance and identify areas for improvement. Additionally, the IMS includes a low-stock alert system that provides automatic notifications when product quantities fall below a defined threshold, helping businesses reorder on time and avoid lost sales. The system also features supplier management tools that assist in maintaining vendor information and tracking purchase orders.

To get started, users can create an account by entering basic information such as name, email, and password. Once registered, they can log in and gain access to the full range of system features. The user-friendly interface ensures that even employees with limited technical experience can easily navigate and benefit from the platform. The IMS is developed using HTML, CSS, and JavaScript for the frontend, while the backend is powered by the Laravel framework, with MySQL serving as the database. A dedicated administrative module is also included, allowing authorized administrators to log in, monitor system activities, manage users, and control all core data.

Overall, the Inventory Management System (IMS) is a digital inventory advisor that enhances business efficiency through centralized tracking, automated alerts, and detailed reporting.

1.2 Problem Statement

Most existing inventory management solutions for small and medium businesses rely on disconnected tools like spreadsheets or simple record-keeping, which are prone to human error and difficult to scale. They often do not offer real-time low-stock alerts or intelligent reporting on sales trends. Also, tracking inventory across different product categories and managing supplier information is inefficient and time-consuming.

1.3 Objectives

- The main objective of this project is to provide businesses with a centralized system for efficient inventory control.
- To support inventory managers by providing real-time tracking of all stock levels and product information.
- To generate automated low-stock alerts and insightful sales reports, helping businesses make informed restocking and sales decisions.
- **To** create a streamlined system for managing supplier details and purchase orders, improving supply chain coordination.

1.4 Scope and Limitation

1.3.1 Scope

- The system provides a centralized product catalog and tracks all inventory movements.
- It offers real-time low-stock alerts and reporting features to support timely business decisions.
- The system includes supplier and purchase order management to coordinate the supply chain.
- The system also includes user management to control access and permission within the application.

1.4.2 Limitation

- The accuracy of stock levels depends on users consistently updating all stock-in and stock-out transactions.
- The system requires internet access for use, which may limit functionality in areas with poor connectivity.
- The initial version of the IMS is designed for a single business or warehouse and does not support multiple, separate legal entities.
  The system cannot automatically predict optimal stock reorder points without historical sales data being sufficiently populated.

1.5 Development Methodology

Waterfall model will be used to build this project. It is also known as a linear-sequential life cycle model. For this project we have clear and fixed requirements, well-understood technology so in order to build this project and this model is easy to understand so, we are using this model to build this project.

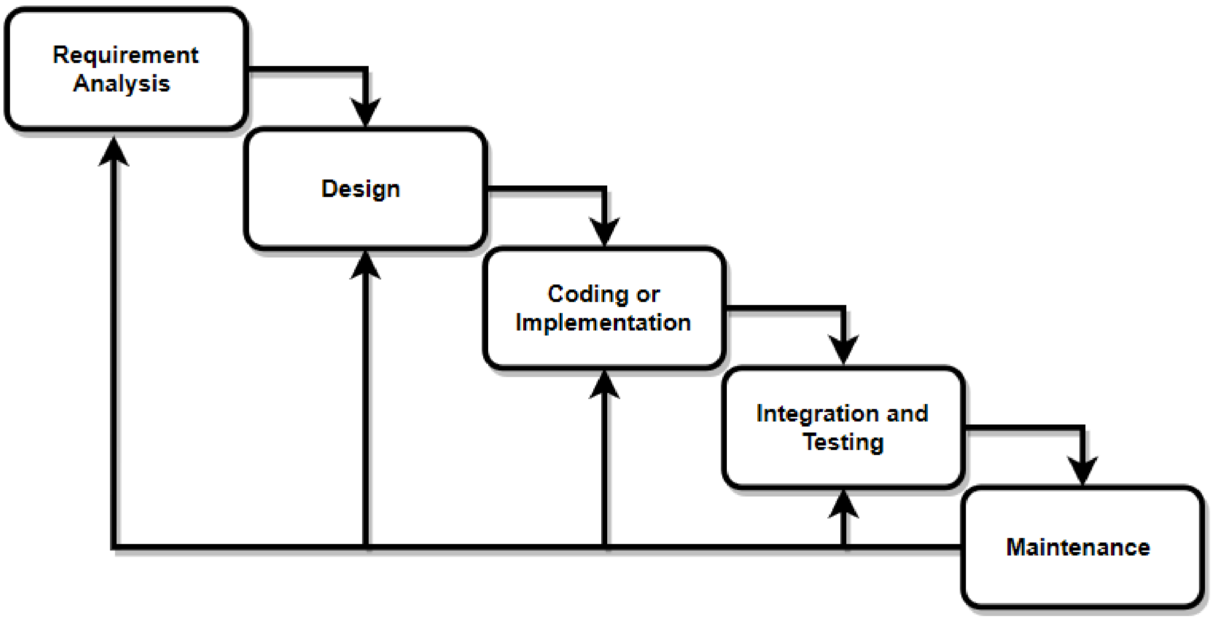

Figure 1.1 Waterfall Model

1.6 Report Organization

The report consists of five chapters in which all the phased of application design and development will be covered.

Chapter 1: The chapter introduces the system and the problems, gives an overview about the study.

Chapter 2: The second chapter covers background study and the literature review of the project.

Chapter 3: The third chapter covers the system analysis and design phase of the application. 3

Chapter 4: The fourth chapter discusses the implementation and testing phase of the application development.

Chapter 5: The last chapter that is the fifth chapter covers the conclusion, recommendations, and future works to improve this project.

CHAPTER 2: BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

The concept of digital inventory management systems like the IMS has gained importance in addressing the challenges faced by modern businesses. Historically, inventory control focused on manual record-keeping and spreadsheets, but today, digital tools aim to provide intelligent support for stock tracking, reporting, and supply chain management. The IMS serves as a centralized platform where businesses can access product information, supplier details, low-stock alerts, and sales reports anytime and anywhere. With the growing need for efficient and data-driven operations, platforms like the IMS promote productivity, reduce risks of overstocking or stockouts, and empower managers through accurate, real-time data. However, challenges such as user adoption, integration with existing systems, and data accuracy still need to be addressed for wider impact.

2.2 Literature Review

Odoo Inventory is an open-source, integrated business management software that includes a comprehensive inventory module. It allows businesses to manage stock levels, track products across multiple warehouses, and automate procurement processes with reordering rules. Its strength lies in its integration with other business functions like sales and purchasing, providing a unified system. [1]

Zoho Inventory is a cloud-based inventory management solution designed to help businesses control their stock and orders. It supports multi-channel sales, order fulfillment, and shipping integration. A key feature is its automation of stock updates across sales channels once an order is placed, helping to maintain accurate inventory counts and prevent overselling. [2]

The Economic Order Quantity (EOQ) model is a fundamental principle in inventory management. It is a formula used to determine the optimal order quantity that minimizes the total holding costs and ordering costs. The model provides a quantitative basis for making restocking decisions, which is a core concept that automated systems like the IMS aim to implement through alerts and reporting. [3]

Enterprise Resource Planning (ERP) systems integrate core business processes, including inventory management, into a single system. The inventory module within an ERP is crucial for providing real-time data across the organization, ensuring that sales, warehouse, and procurement departments all work with the same information, thus improving coordination and efficiency. [4]

inFlow Inventory is a desktop and web-based inventory management software designed for small to medium-sized businesses. It focuses on user-friendliness and core features such as stock tracking, barcode scanning, and generating basic purchase orders and sales reports, making it a practical solution for companies transitioning from manual methods. [5]

TradeGecko (now QuickBooks Commerce) is a cloud-based inventory and order management platform built for modern brands and wholesalers. It is designed to centralize operations across sales channels, manage complex inventories, and automate order workflows, helping businesses scale their operations efficiently by providing a clear view of their entire supply chain. [6]

Fishbowl Inventory is a powerful manufacturing and warehouse management system (WMS) that integrates with QuickBooks. It is designed for small to medium-sized businesses that require advanced features like asset tracking, work orders, and bill of materials (BOM) management, offering a robust solution for complex inventory needs beyond simple stock counting. [7]

Just-In-Time (JIT) Inventory is a management philosophy that aligns raw-material orders from suppliers directly with production schedules. Companies use this strategy to increase efficiency and decrease waste by receiving goods only as they are needed in the production process, thereby reducing inventory costs. This method requires accurate inventory tracking and reliable suppliers. [8]

Lightspeed Retail is a point-of-sale and e-commerce platform that includes robust inventory management features. It is tailored for retail businesses, allowing them to manage stock across multiple physical store locations and an online shop from a single dashboard, syncing data in real-time to provide a unified view of sales and inventory. [9]

CHAPTER 3: SYSTE M ANALYSIS AND DESIGN

3.1 System Analysis

During the system analysis phase, the project's components were examined to identify inefficiencies and ensure alignment with the objectives. Workflows, database structures, and user interactions were analyzed. Key challenges were identified and addressed, resulting in a more efficient system.

3.1.1 Requirement Analysis

The requirement analysis of the IMS is done through finding the functional requirements and non-functional requirements for the system.

Functional Requirements

Use case diagrams were developed in order to identify functional requirements, which aided in identifying the IMS's key features. These features were outlined to make sure the system helps users effectively accomplish their objectives.

Use Case Diagram

The "Inventory Management System" consists of three actors: Admin, Manager, and User.

Admin: The Admin shall log in to the system. The Admin shall manage all product categories, supplier information, and product details. The Admin shall also be able to view all users, and add or remove Managers.

Manager: The Manager shall log in to the system. The Manager can perform all inventory tasks, such as managing the product catalog, updating stock levels, viewing reports, and handling low-stock alerts. Furthermore, the Manager shall be able to add or remove other Managers and Users.

User: The User shall log in to the system. The User can view the product, check stock levels and perform purchase and sale activities while after the purchase order is created the Admin and the Manager have the right to approve the purchase order. The User does not have any user management permissions.

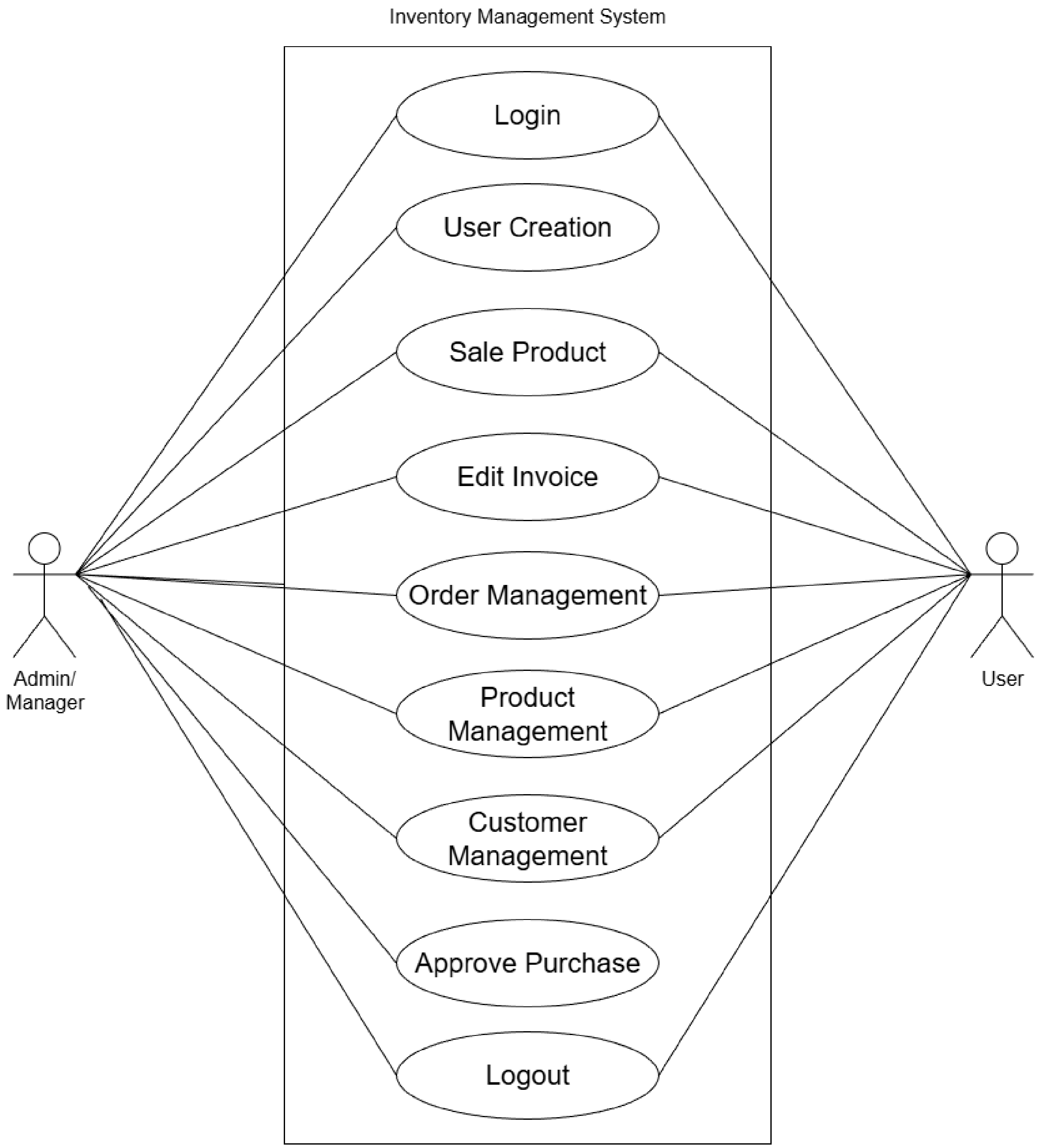

Figure 3.1: Use Case Diagram

Non-Functional Requirements

These attributes were considered critical in enhancing the overall performance and usability of the IMS. Key parameters to ensure non-functional requirements are:

- **Performance:** The system is able to handle a large amount of inventory data and transaction processing efficiently.
- **Security:** Security is enhanced by restricting system changes to administrators only and encrypting sensitive details like passwords. Each user has a unique session upon logging in, requiring their login details to access the system securely.
- **Usability:** The system is user friendly and easy to use and understand, with a clear and intuitive interface.
- **Accessibility:** The system is accessible to authorized users via a web browser, regardless of their location.
- **Maintainability:** The system is easy to maintain and update.

3.1.2 Feasibility Analysis

We conducted a detailed feasibility analysis to assess the potential success of the proposed Inventory Management System. This analysis helped us evaluate whether the project is practical, achievable, and beneficial from various perspectives including technical, operational. economic, and schedule feasibility

The four key aspects of feasibility considered are as follows:

i. Technical Feasibility

The IMS is technically feasible within a college project context. Using Laravel for the backend and HTML, CSS, and JavaScript for the frontend allows for rapid development, efficient data handling, and a clean separation between the business logic and user interface, resulting in a smooth and scalable web application experience.

ii. Operational Feasibility

The IMS is operationally feasible, offering a user-friendly interface that allows employees with basic computer skills to use the system effectively. The system supports essential features like product management, stock tracking, and reporting, making it practical and manageable for regular use without requiring advanced technical knowledge.

iii. Economic Feasibility

Economic feasibility was examined to assess the financial viability of the project. Development costs were estimated, focusing on freely available tools and resources which showed that the project is economically feasible.

iv. Schedule Feasibility

Schedule feasibility was assessed to determine whether the project timeline and deadlines were reasonable. Consideration was given to the constraints of limited time, ensuring that the project could be completed within the specified time frame, even under mandatory deadlines.

Figure 3.2: Gantt Chart of IMS

3.1.3 Object Modelling

i. Object Diagram

The Object Diagram represents a snapshot of the Inventory Management System at a specific point in time, showing concrete instances of the classes with real data values. Unlike the class diagram, which is abstract, the object diagram focuses on the runtime state of the system.

For example:

- A User instance, *Nikesh (Manager)*, with user ID 1, interacts with the system.
- His role instance is Manager, defined in the Role class.
- A Product instance, *Laptop (P01)*, with a stock quantity of 10 and price of 80,000, is part of the inventory.
- A Sale instance S001 represents a transaction on 2025-08-30 with a total amount of 160,000.
- The SaleDetail instance D001 links the sale with the product, showing that two laptops were sold in this transaction.

This diagram makes it easier to visualize real-world interactions within the system, demonstrating how users, products, and sales data are connected during actual execution.

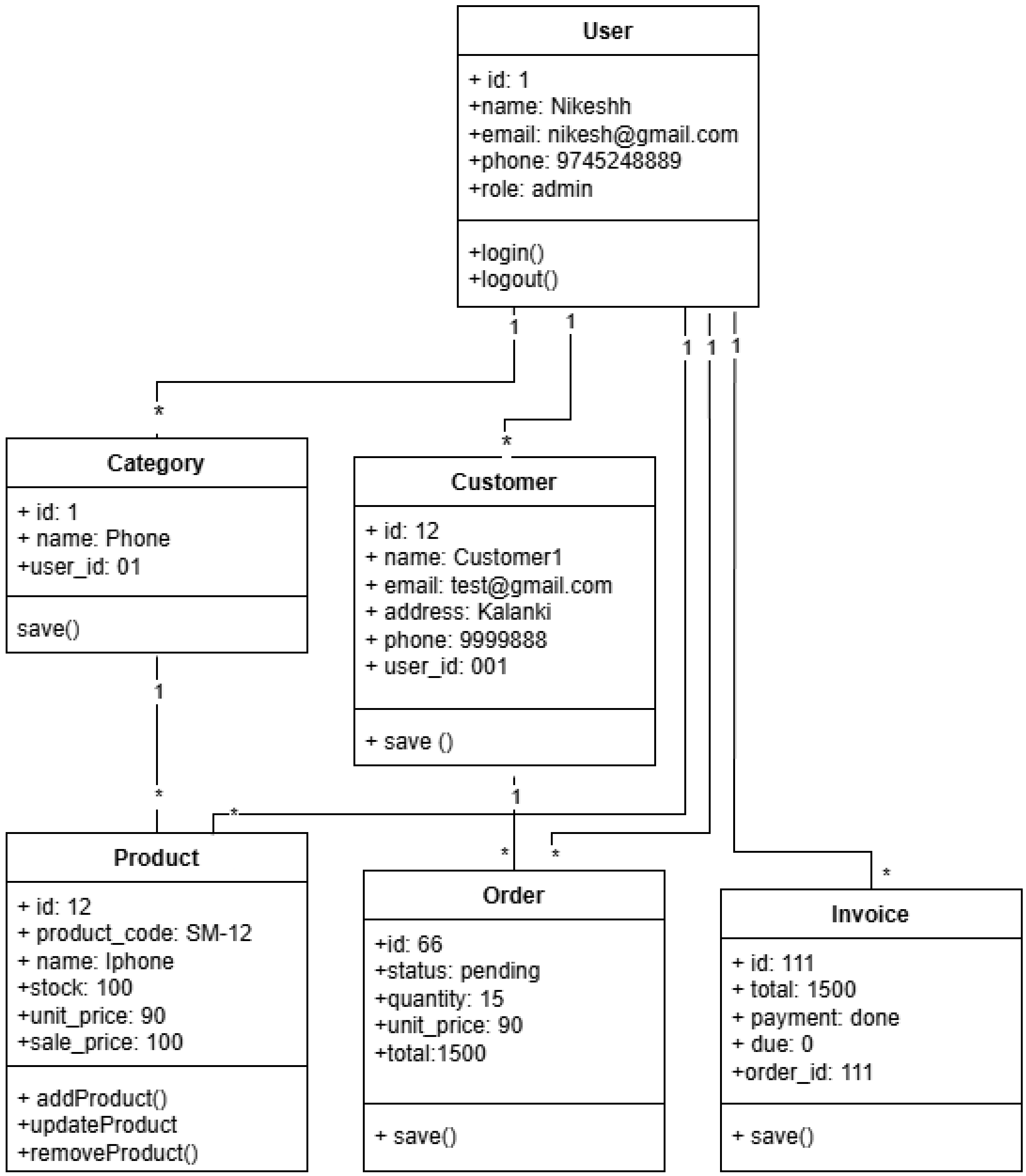

Figure 3.3: Object Diagram of Inventory Management System

ii. Class Diagram

The class diagram represents the structure and relationships within the database models of the IMS, including the differentiation of user roles and relationships among various classes.

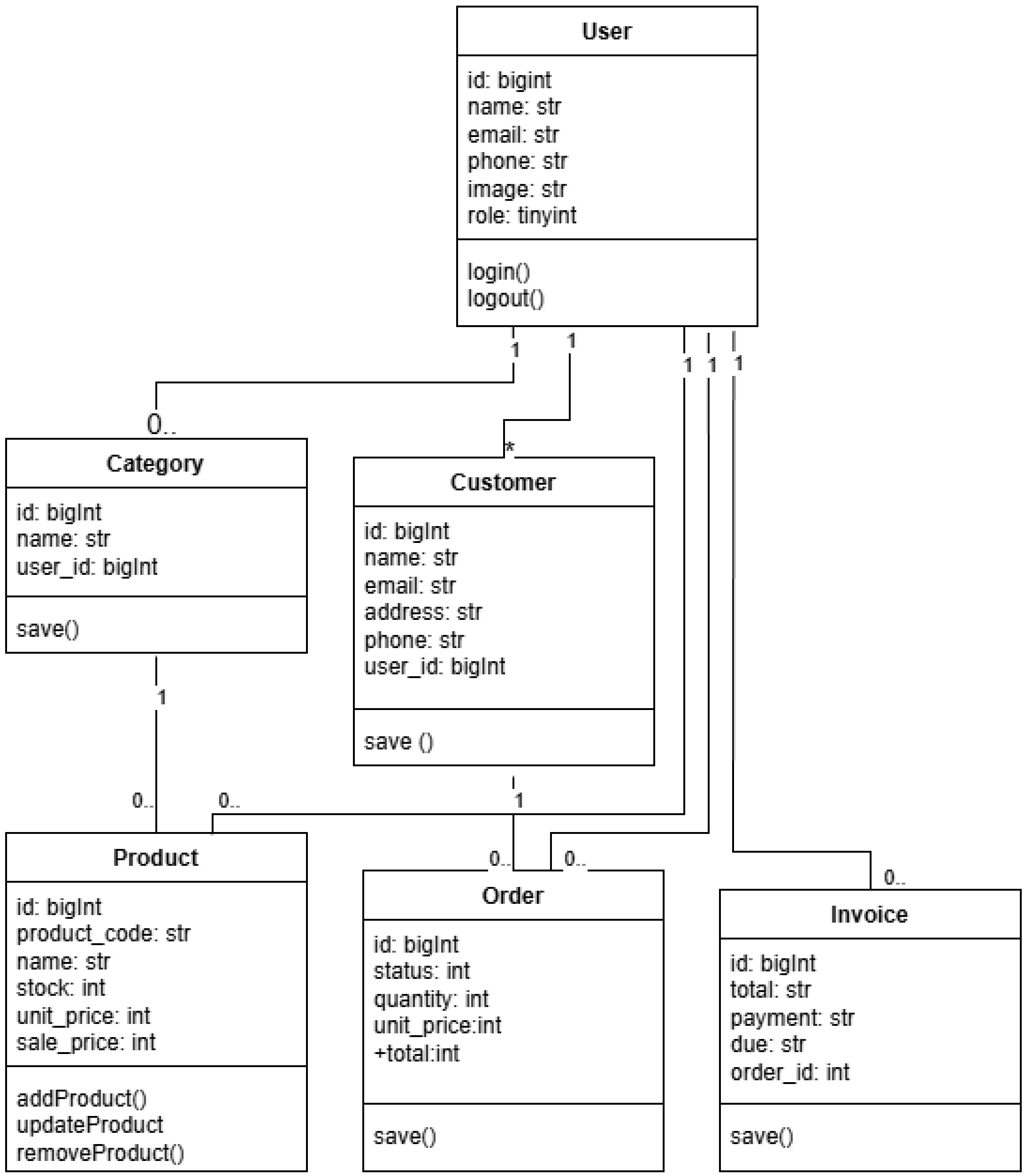

Figure 3.4: Class Diagram of Inventory Management System

3.1.4. Dynamic Modeling

i. State Diagram

The following state diagram demonstrates different lifecycle states that the entries go through.

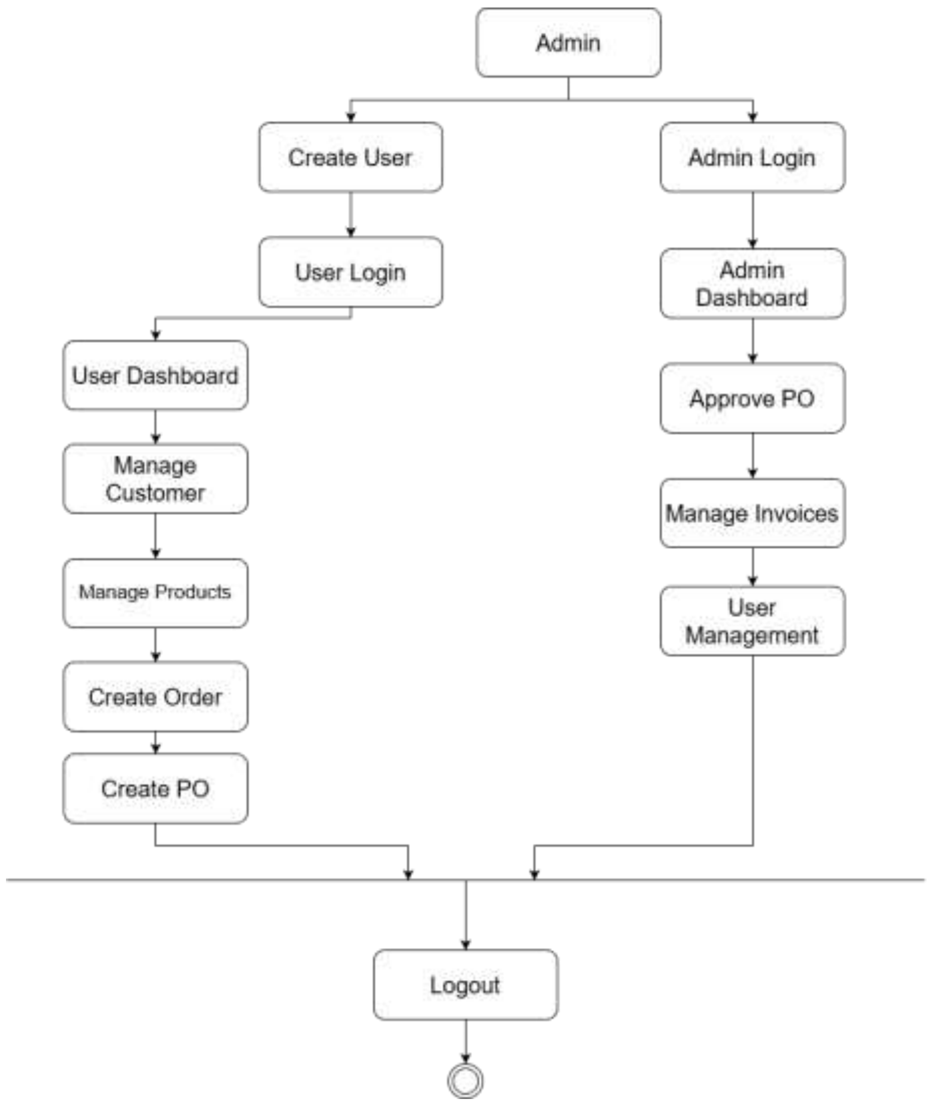

Figure 3.5: State Diagram of Inventory Management System

ii. Sequence Diagram

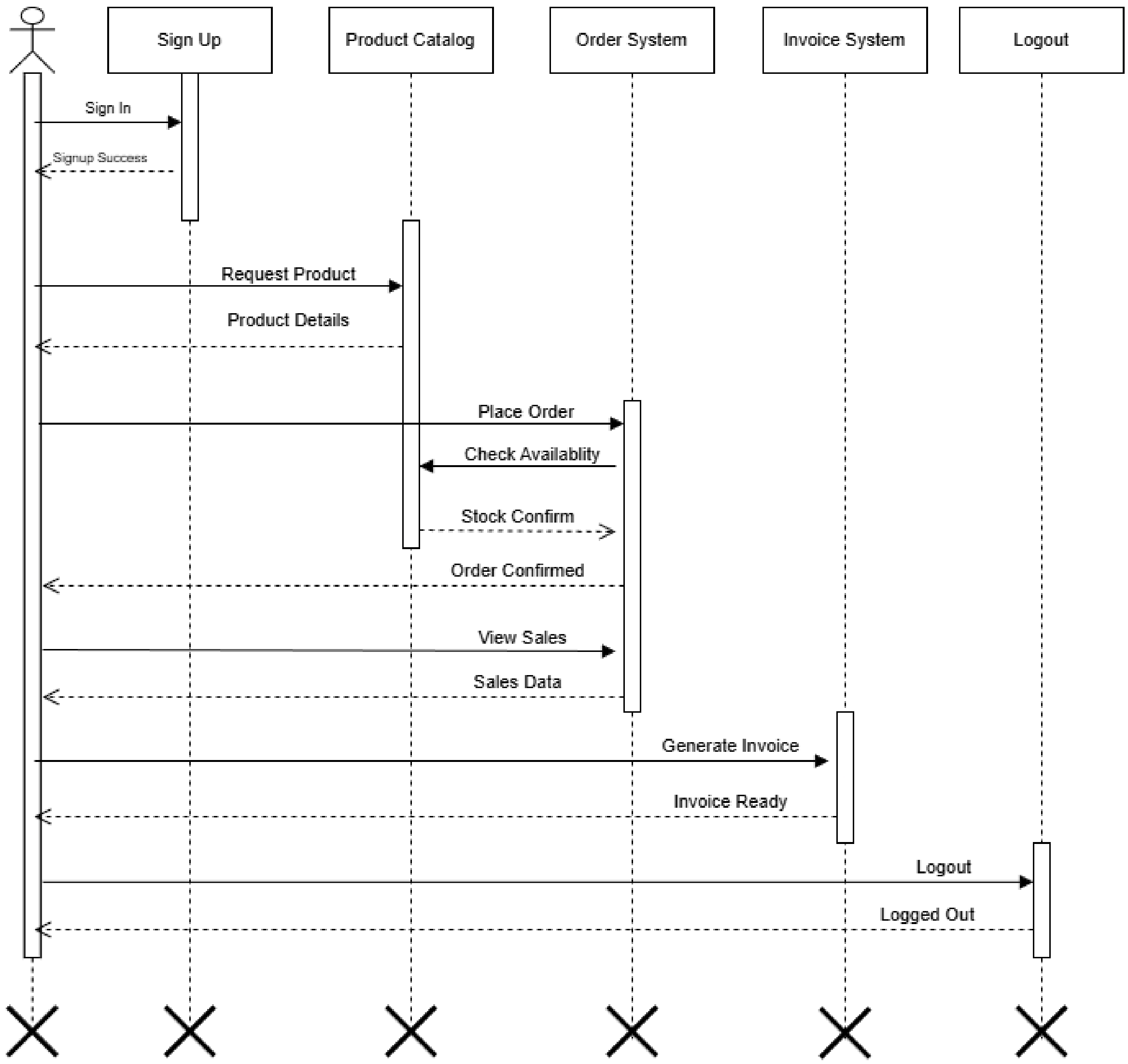

Figure 3.6: Sequence Diagram of User in IMS

Figure 3.6 shows the sequence diagram of User where the user would sign up or log in to the system to access features such as browsing products, placing orders, viewing order history, and managing their profile.

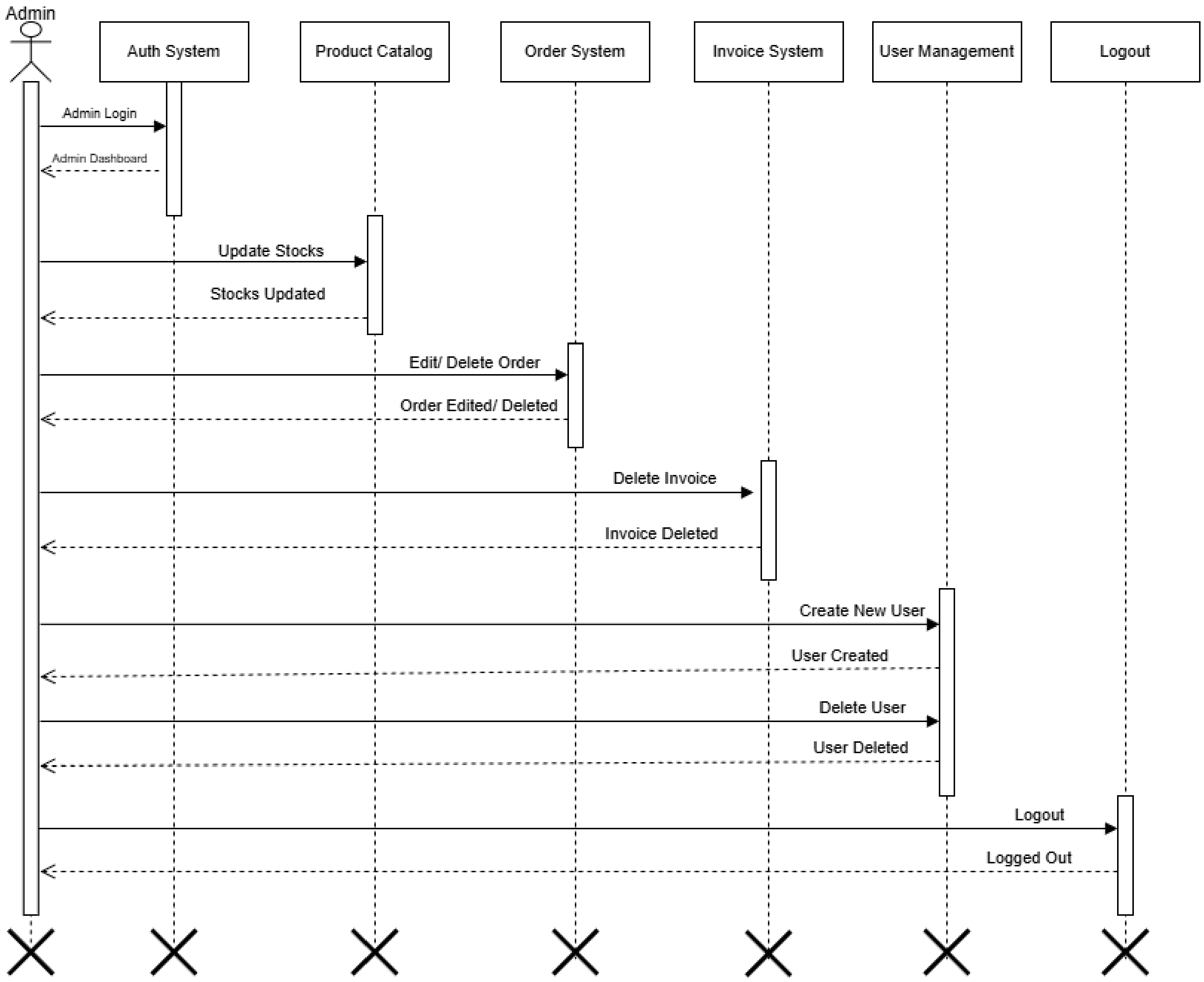

Figure 3.7: Sequence Diagram of Admin in IMS

Figure 3.7 shows the sequence diagram of Admin where the admin would log in to the system to access administrative features such as managing products, processing orders, monitoring inventory levels, generating reports, and overseeing system operations.

3.1.5. Process Modeling

i. Activity Diagram for Admin

The Admin Activity Diagram illustrates the systematic workflow for inventory system administrators. The process initiates with secure login authentication, leading to dashboard access. Administrators sequentially perform product management, order processing, and report generation tasks. The workflow emphasizes systematic inventory control through product catalog maintenance, sales order management, and business intelligence reporting, concluding with secure session termination.

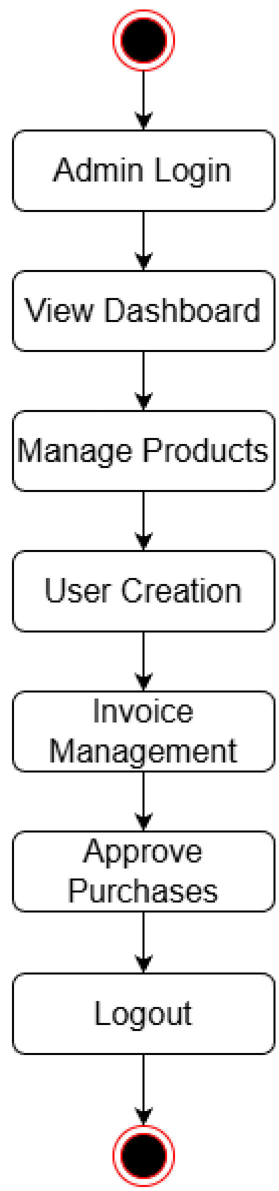

Figure 3.8: Activity Diagram for Admin

ii. Activity Diagram for User

The User Activity Diagram demonstrates the customer journey within the inventory management system. Starting with authentication, users proceed to browse available products, place orders, and monitor their purchase history. The workflow emphasizes user-friendly navigation through product discovery, streamlined ordering processes, and transparent order tracking capabilities. The session concludes with secure logout, ensuring data privacy and system security.

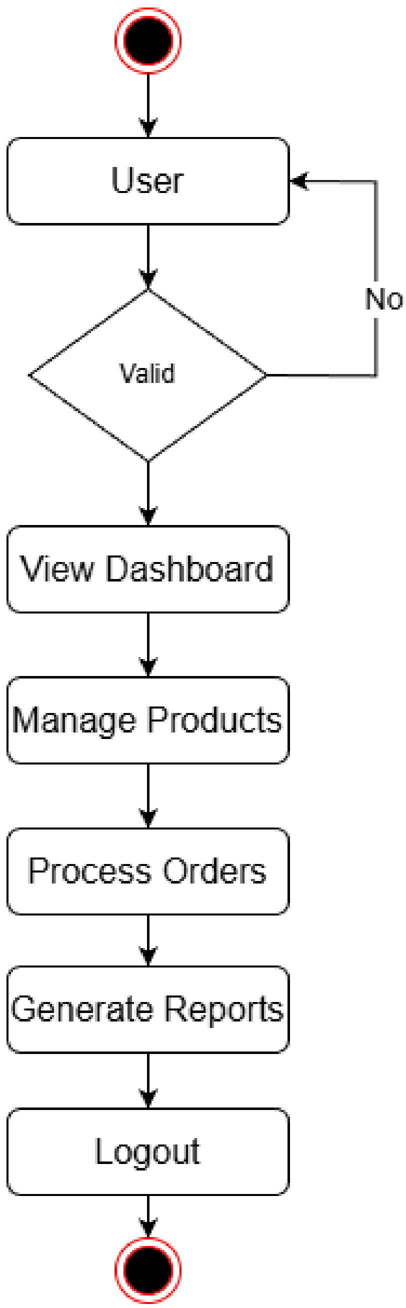

Figure 3.9: Activity Diagram for User

3.2 System Design

During the system design phase, the analysis is taken to another step where existing classes and object diagram are refined and component and deployment diagram are constructed.

3.2.1 Refinement of Class Diagram

In the following refined diagram, the existing class diagram has been refined by adding more detail. The diagram consists of method signatures, return types and parameter types. While the class diagram showed the overview of system, the refined class diagram adds more detail for actual implementation.

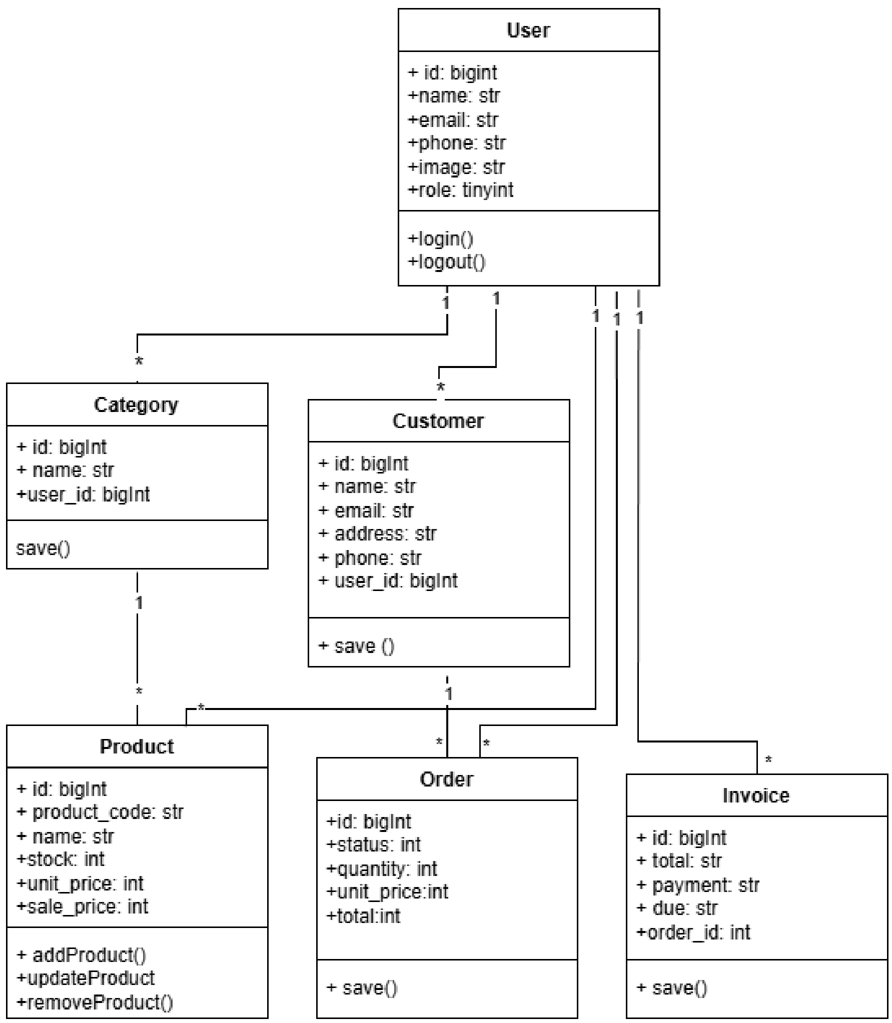

Figure 3.10: Refined Class Diagram

3.2.2 Component Diagram

The component diagram below illustrates the physical components of the Inventory Management System and their interactions. With its help we can visualize the overall structure and organization of the system.

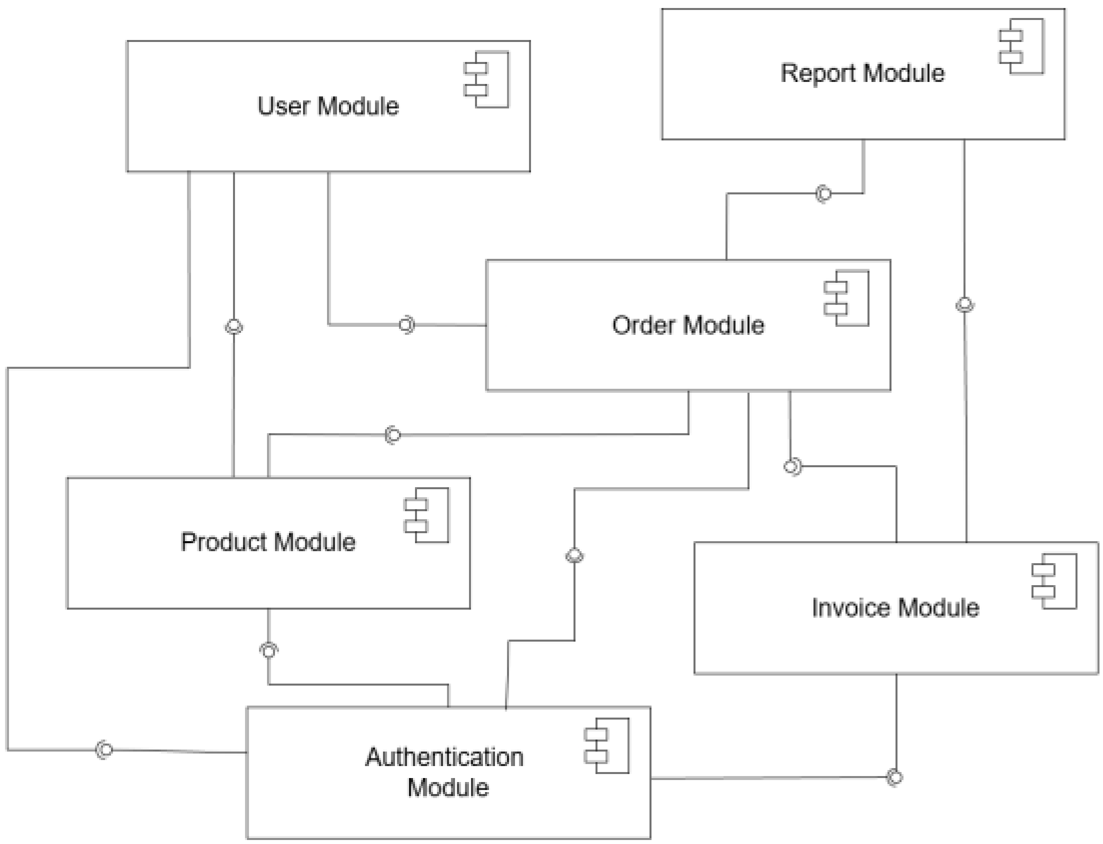

Figure 3.11: Component Diagram of IMS

3.2.3 Deployment Diagram

The deployment diagram illustrates the physical deployment of the Inventory Management System across various hardware nodes and execution environments. It shows how the system components are distributed across different servers and devices.

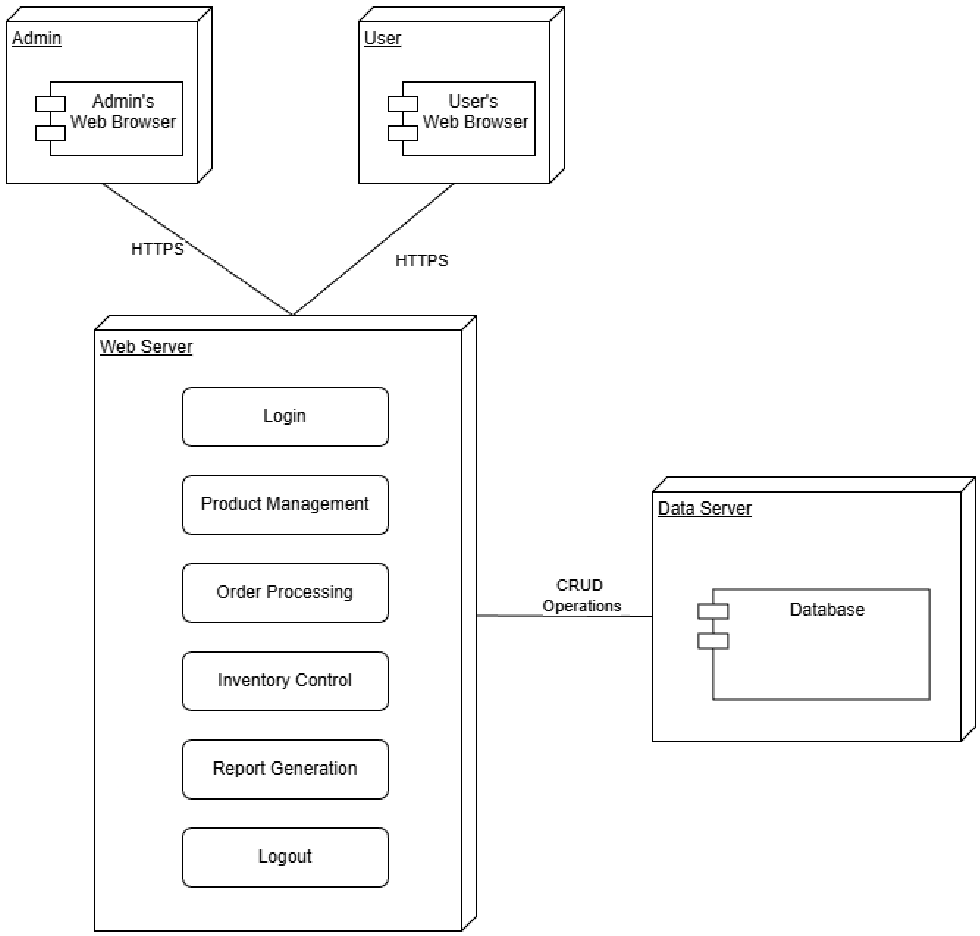

Figure 3.12: Deployment Diagram of IMS

3.3 Algorithm Details

In the Inventory Management System (IMS), the K-Means Clustering algorithm has been implemented to analyze and group products based on their sales performance. This helps the system automatically identify high-selling, medium-selling, and low-selling products. Such analysis assists managers and administrators in making better decisions regarding stock management, discounts, and restocking priorities.

K-Means Clustering Algorithm

K-Means is an unsupervised machine learning algorithm used to group similar data points into clusters based on their numerical characteristics. In the IMS, the data points represent products, and features such as sales count, revenue, or frequency of purchase are used for clustering.

Mathematical Model

The algorithm aims to minimize the sum of squared distances between each data point and its assigned cluster center (centroid).

$$J = \sum_{i=1}^{k} \sum_{x_j \in C_i} \| x_j - \mu_i \|^2$$

Where:

- k = number of clusters
- x_j = data point (e.g., product's sales)
- μ_i = centroid of cluster i
- C_i = set of data points in cluster i

Steps

1. **Load Dataset:** Collect product sales data from the system database. Each record includes product ID, name, total sales, and revenue.
2. **Select Number of Clusters (k):** Decide how many clusters to create (for example, 3 clusters → High, Medium, and Low sales).
3. **Initialize Centroids:** Randomly select k products as the initial cluster centroids.

4. **Assign Products to Nearest Centroid:** For each product, calculate the distance between its sales value and all centroids using Euclidean distance: $D = \sqrt{} (x - \mu i)^2$
   Assign the product to the cluster with the nearest centroid.
5. **Update Centroids:** Recalculate each centroid as the average sales of all products assigned to that cluster.

$$\mu_i = \frac{1}{n_i} \sum_{x_j \in C_i} x_j$$

6. Repeat: Repeat steps 4 and 5 until centroids stop changing or changes become minimal.
7. Store Cluster Results:
   The final clusters represent groups of products:
   - Cluster 1: High-selling products
   - Cluster 2: Medium-selling products
   - Cluster 3: Low-selling products

Stopping Condition

The algorithm stops when:

- Centroid positions remain unchanged between iterations, or
- The number of iterations reaches the defined maximum limit.

Prediction / Group Assignment

When a new product's sales data is recorded, the system calculates its distance to each cluster centroid and assigns it to the nearest cluster. This allows automatic categorization of new products into appropriate sales groups.

Role in Inventory Management System

- Helps managers analyze sales trends.
- Supports data-driven restocking decisions.
- Assists in promotional planning by identifying low-selling items.
- Enhances report generation by grouping similar products together.

CHAPTER 4: IMPLEMENTATION AND TESTING

4.1 Implementation

4.1.1 Tools Used

Laravel

Laravel powers the backend services in the Inventory Management System, enabling the development of scalable and secure web applications with minimal setup. It handles business logic, data processing, and integrates seamlessly with MySQL. With built-in tools for authentication, validation, and performance monitoring, Laravel ensures a reliable and maintainable backend architecture.

Blade Templating

Blade is used as the server-side templating engine in the IMS to dynamically generate HTML pages. It seamlessly integrates with Laravel to render user interfaces using real-time data. Blade ensures a clean separation between logic and design, making the platform maintainable and user-friendly.

HTML, CSS and JavaScript

HTML is used to build all the basic structure of this project. HTML is used for the front-end structure. CSS is used for styling of the web pages, including colors, layout and fonts. It is a simple design language intended to simplify the process of making web pages presentable. JavaScript is used for the validation of forms and ensuring the responsiveness of web pages. It is used to make web pages interactive, allowing for real-time user interactions and enhancing the overall user experience.

MySQL

MySQL serves as the database for our system, storing and managing critical data such as user profiles, product information, customer details, and transaction records. Known for its reliability and support for complex queries, MySQL ensures efficient data storage and retrieval, while maintaining data integrity and scalability for our platform.

4.1.2 Implementation Details of Modules

The major functional modules of the Inventory Management System and their implementation details are explained below.

- **Register Module**: Enables new users to create accounts, collecting necessary information for user profiles, and allowing access to system resources.

- **Product Management Module**: Allows admin to add, edit, and manage product information including categories, prices, and stock levels.
- **Order Processing Module**: Enables users to place orders and allows administrators to process, track, and manage order status throughout the fulfillment cycle.
- **Inventory Tracking Module**: Provides real-time stock level monitoring, low-stock alerts, and inventory status updates for efficient stock management.
- **Reporting Module**: Generates sales reports, inventory analytics, and business insights to support data-driven decision making.
- **Login Module** Allows users to securely log into their accounts, providing access to personalized content and features according to their roles.

4.2 Testing

In this step, the system is evaluated to test if it performs as expected. Testing in the Inventory Management System is done by testing every unit and module.

We mainly focused on the **functionality**, **usability**, and **basic security** of the system. Some key modules like login, product management, stock updates, and order creation were tested with different types of inputs to make sure they behave properly.

Test cases were written and manually checked to see if expected outputs were being returned. We also tested for edge cases like empty fields, invalid inputs, and unauthorized access.

To check cross-browser compatibility, we ran the system on multiple browsers like:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Brave

4.2.1 Test Case for Unit Testing

Unit testing focuses on the smallest parts of the system like individual functions, routes, or components to ensure they behave as expected. These tests were mostly done manually and a few through Laravel's built-in testing framework **(php artisan test).**

We tested login, customer registration, product handling, and order functionality. Below is the list of some key test cases performed

Table 4.1 Test Cases for Registration

Project Name: Inventory Management System					
Test Case 1					
Test Case ID: TC-1			**Test Designed by: Nikesh Manandhar**		
Module Name: Register			**Test Executed by: Nikesh Manandhar**		
Description: Testing the User Registration functionality of Inventory Management System					
S.N.	Test Case	Input	Expected Result	Actual Result	Status
1	Valid Registration	name="Nikeshh", email="nikeshh@test.com", password="nikeshh@123", confirm_password="Admin@123"	Registration Success	Registration Success	Pass
2	Password Mismatch	password=" nikeshh@123", confirm_password=" nikeshh@124"	Registration Fail	Registration Failed	Pass
3	Duplicate Email	email="admin@ims.com" (already exists)	Registration Failed	Registration Failed	Pass
4	Valid Phone Format	Phone = "9745248880"	Registration Success	Registration Success	Pass
5	Invalid Email Format	email: "invalid-email"	Registration Fail	Registration Fail	Pass
6	Valid Email	Email: "nikeshh@gmail.com"	Registration Success	Registration Success	Pass

Table 4.2 Test Case for Login

Project Name: Inventory Management System					
Test Case 2					
Test Case ID: TC-2			**Test Designed by: Nikesh Manandhar**		
Module Name: Login			**Test Executed by: Sudip Magar**		
Description: Testing the User Login functionality					
S. N	**Test Case**	**Input**	**Expected Result**	**Actual Result**	**Status**

1	Valid Login	email: nikesh@test.com	Login Success	Login Success, Dashboard Loaded	Pass
2	Invalid Route	127.0.0.1:8000/xyz	"404 Page Not Found"	404 Not Found	Pass
3	Invalid Credentials	Email: wrong@test.com Password: “wrongpass”	Login Failed	Login Failed	Pass
4	Empty Credentials	Email: “” Password: “”	Login Failed	Login Failed	Pass

Table 4.3 Test Case for Order Module

Project Name: Inventory Management System					
Test Case 3					
Test Case ID: TC-3			**Test Designed by: Sudip Magar**		
Module Name: Order			**Test Executed by: Nikesh Manandhar**		
Description: Testing the Order functionality					
S.N	**Test Case**	**Input**	**Expected Result**	**Actual Result**	**Status**
1	Create Valid Order	product_id=1, customer_id=1, quantity=2	Order Created Successfully	Order Created Successfully	Pass
2	Insufficient Stock	quantity=1000 (stock=50)	"Insufficient stock"	“Insufficient Stock”	Pass
3	Invalid Quantity	quantity=0 or -5	"Invalid quantity"	“Invalid quantity”	Pass
4	Update Status	status="processing"	Status Updated	Status Updated	Pass
5	Cancel Order	status="cancelled"	Order Cancelled, Stock Restored	Order Cancelled and Stock Restocked	Pass
6	Price Calculation	quantity=2, price=100	Total=200	Total = 200	Pass

Table 4.4 Test Case for Customer Module

Project Name: Inventory Management System					
Test Case 4					
Test Case ID: TC-4			**Test Designed by: Nikesh Manandhar**		
Module Name: Customer			**Test Designed by: Sudip Magar**		
S.N	**Test Case**	**Input**	**Expected Result**	**Actual Result**	**Status**
1	Add Customer	name="John", email="john@test.com"	Customer Added	Customer Added Successfully	Pass
2	Update Customer	customer_id=1, phone="9841234567"	Customer Updated	Customer Updated Successfully	Pass
3	Delete Customer	customer_id=1	Customer Removed	Customer Removed from System	Pass
4	Duplicate Email	email="admin@ims.com" (exists)	"Email exists"	"Email already registered"	Pass
5	Invalid Email	email="invalid-email"	"Invalid email"	"Please enter valid email"	Pass

Table 4.5 Product and Category Management Test Case

Project Name: Inventory Management System					
Test Case 5					
Test Case ID: TC-5			**Test Designed by: Nikesh Manandhar**		
Module Name: Product & Category			**Test Executed by: Nikesh Manandhar**		
Description: Product and Category Module Test Case					
S.N	**Test Case**	**Input**	**Expected Result**	**Actual Result**	**Status**
1	Add Category	name="Electronics"	Category Added	Category Added Successfully	Pass
2	Update Category	category_id=1, name="Mobile"	Category Updated	Category Updated Successfully	Pass

3	Delete Category	category_id=1	Category Removed	Category Removed from System	Pass
4	Duplicate Name	name="Mobile" (exists)	"Name exists"	"Category name already exists"	Pass
5	Add Category	name="Electronics"	Category Added	Category Added Successfully	Pass
6	Update Category	category_id=1, name="Mobile"	Category Updated	Category Updated Successfully	Pass
7	Delete Category	category_id=1	Category Removed	Category Removed from System	Pass
8	Duplicate Name	name="Mobile" (exists)	"Name exists"	"Category name already exists"	Pass

Table 4.6 Sales Module Test Cases

Project Name: Inventory Management System					
Test Case 6					
Test Case ID: TC-6			**Test Designed by: Sudip Magar**		
Module Name: Sales & Invoice			**Test Executed by: Nikesh Manandhar**		
Description: Sales & Invoice Module Test Cases					
S.N	**Test Case**	**Input**	**Expected Result**	**Actual Result**	**Status**
1	Generate Report	date_range="2024-01-01 to 2024-01-31"	Sales Report	Report Generated Successfully	Pass
2	Top Products	limit=10	Top 10 Products	Top Products List Displayed	Pass
3	Identify fast, slow selling products	Product Sold Highest = Fast Selling	Categorized in Fast	Categorized in Fast	Pass
4	Edit Invoice	Change the Amount	Cannot change the Invoice	Edit Disabled	Pass

5	Generate Invoice	order_id=1	Invoice Created	Invoice Generated Successfully	Pass
6	Update Payment	invoice_id=1, amount=500	Payment Updated	Payment Recorded Successfully	Pass
7	View Invoice	invoice_id=1	Invoice Details	Invoice Details Displayed	Pass
8	Calculate Due	total=200, paid=150	Due=50	Due Amount Calculated Correctly	Pass

CHAPTER 5: CONCLUSION AND FUTURE RECOMMENDATIONS

5.1 Conclusion

The Inventory Management System demonstrates how modern technology can be effectively integrated into business operations to empower managers with better inventory control and decision-making tools. By providing real-time stock tracking, automated reporting, and streamlined order processing, the system enhances operational efficiency and reduces manual errors. Its user-friendly design ensures accessibility for all staff members, regardless of their technical background. Overall, the IMS bridges the gap between traditional inventory methods and digital innovation, paving the way for smarter and more efficient business management.

5.2 Lesson Learnt / Outcome

Developing the Inventory Management System was a valuable experience that taught many important lessons. One key takeaway was the importance of focusing on the users. Understanding what business managers and inventory staff need and prefer was crucial for creating an easy-to-use platform. Building a system that can scale and perform well was also essential. Using technologies like HTML, CSS and JavaScript for the frontend, Laravel for the backend, and MySQL for the database allowed the platform to handle more data and users without slowing down. Additionally, working on database design, REST API creation, and comprehensive testing provided valuable hands-on experience. These lessons will be very useful for future software development projects, helping to create innovative and effective business applications.

5.3 Future Recommendations

Looking ahead, the Inventory Management System can be enhanced by integrating emerging technologies such as AI and machine learning to provide predictive analytics for demand forecasting and automated reordering. Incorporating barcode and QR code scanning capabilities will streamline product identification and stock counting processes. Expanding mobile optimization, especially for offline access, will make the platform more accessible to warehouse staff in areas with limited connectivity. Integrating IoT devices like smart shelves and inventory drones can further support real-time stock monitoring. Strengthening data security measures and offering role-based access control can build user trust. Additionally, creating advanced analytics features that provide deeper business insights and integration with accounting software will further align the system with comprehensive business needs. Lastly, providing multi-warehouse support and supply chain integration will make the platform more robust for growing businesses.

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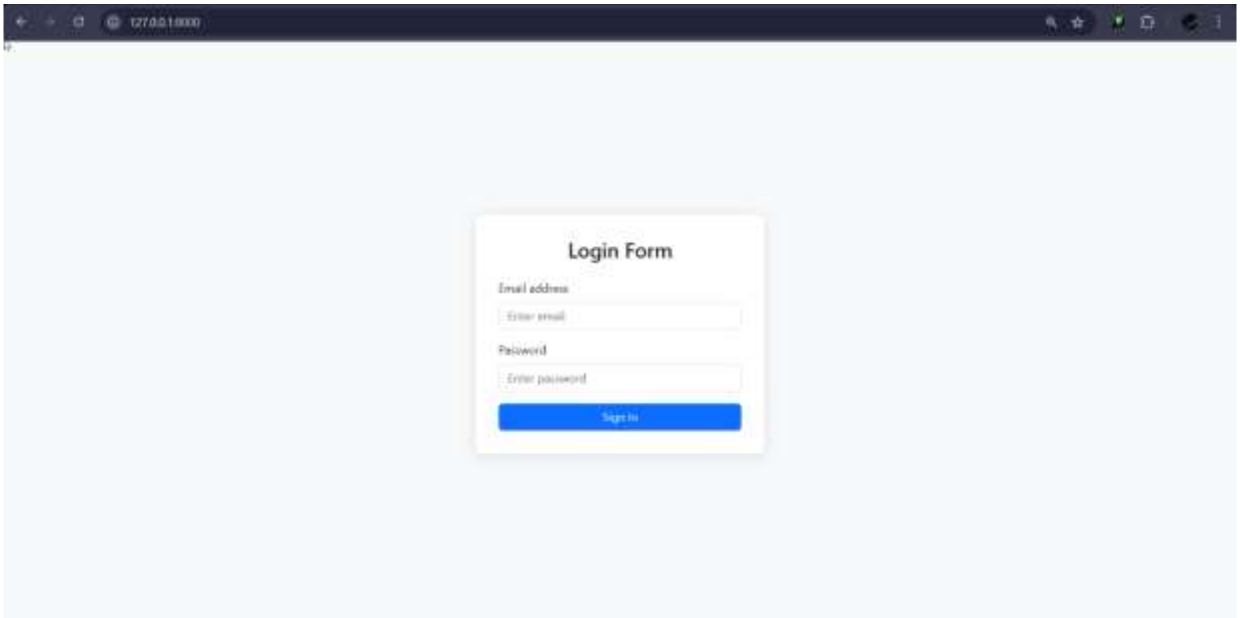
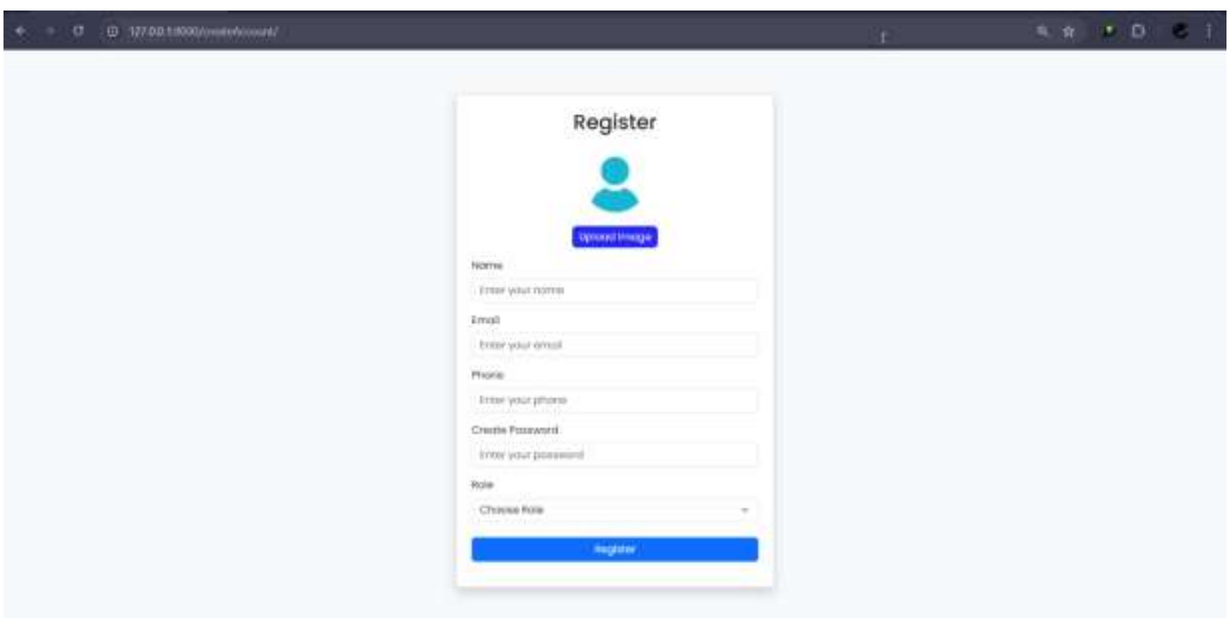
APPENDICES

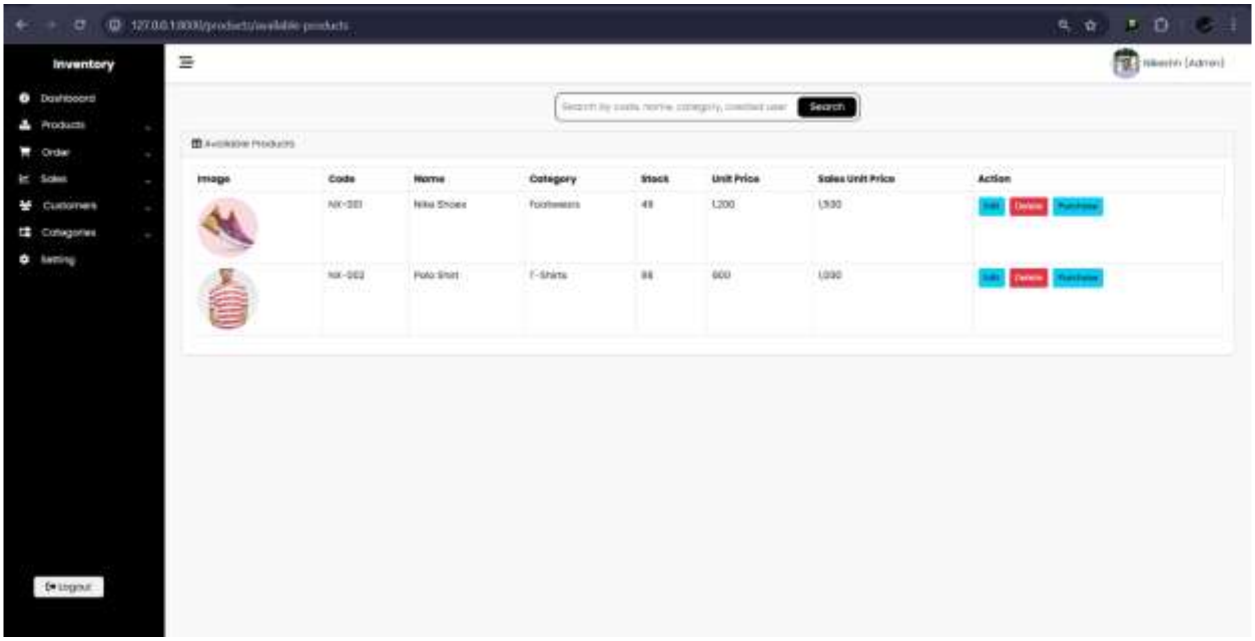
Appendix 1: Screenshots



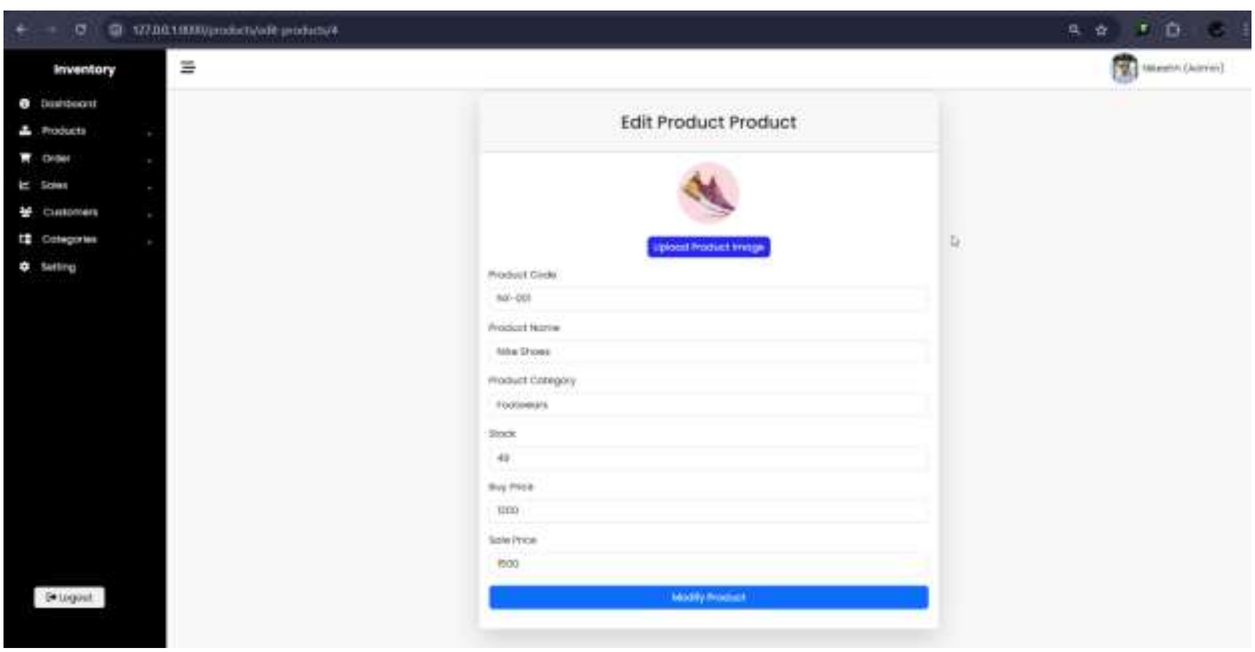
Login Page



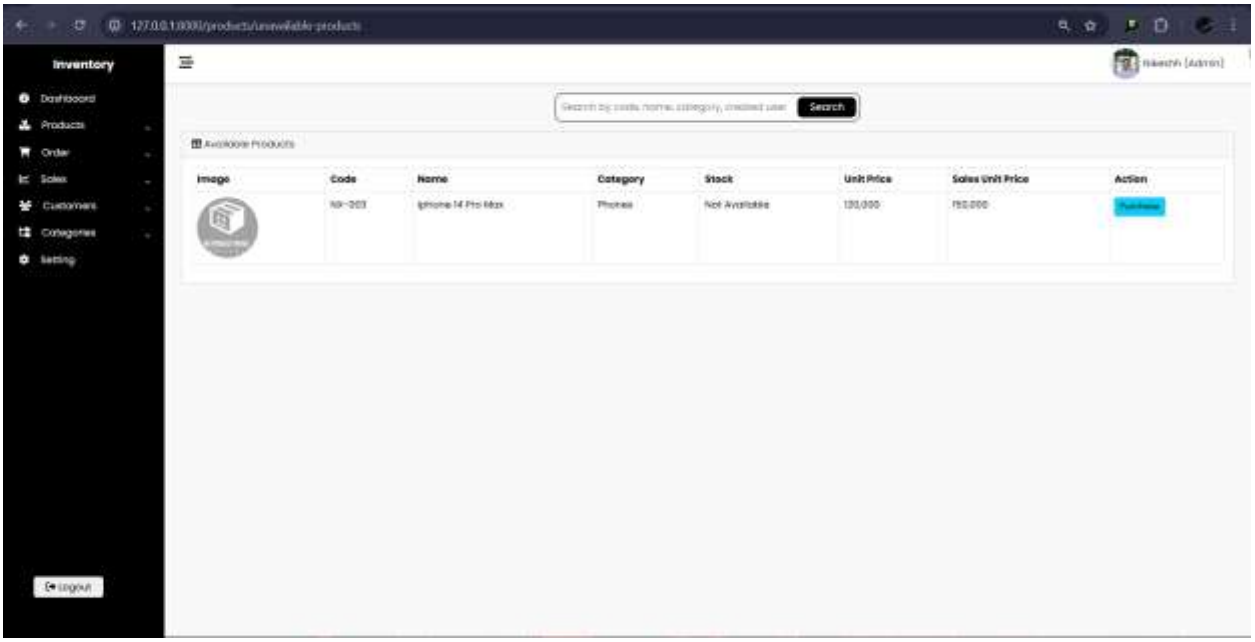
Register Page



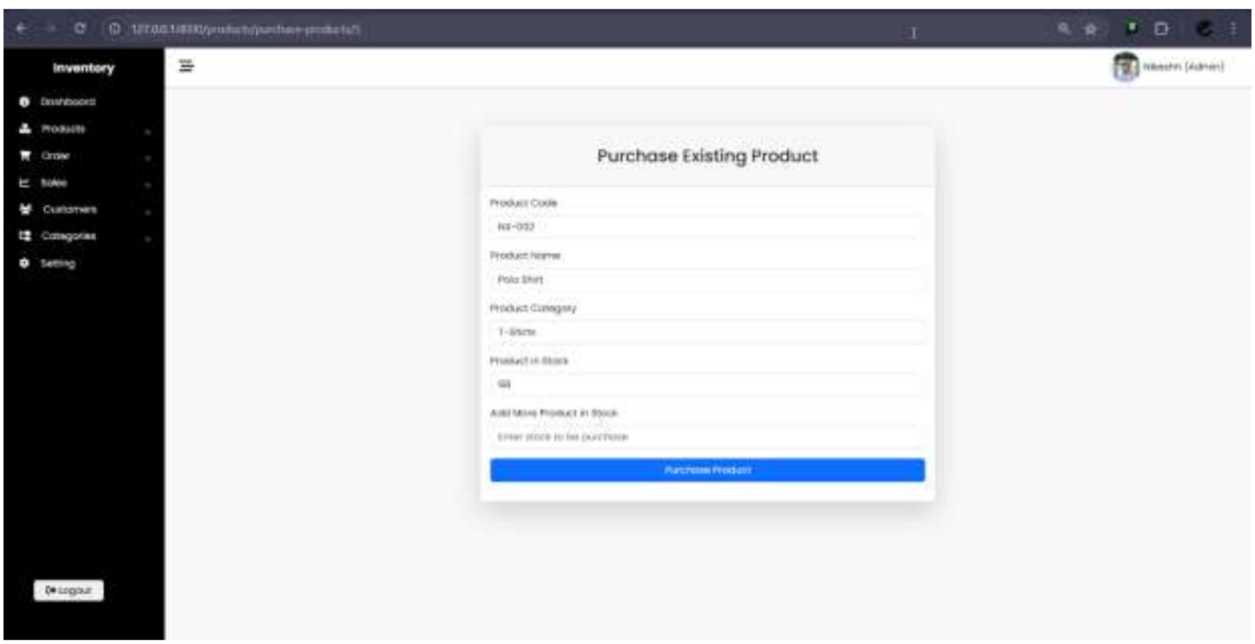
Product Login Page



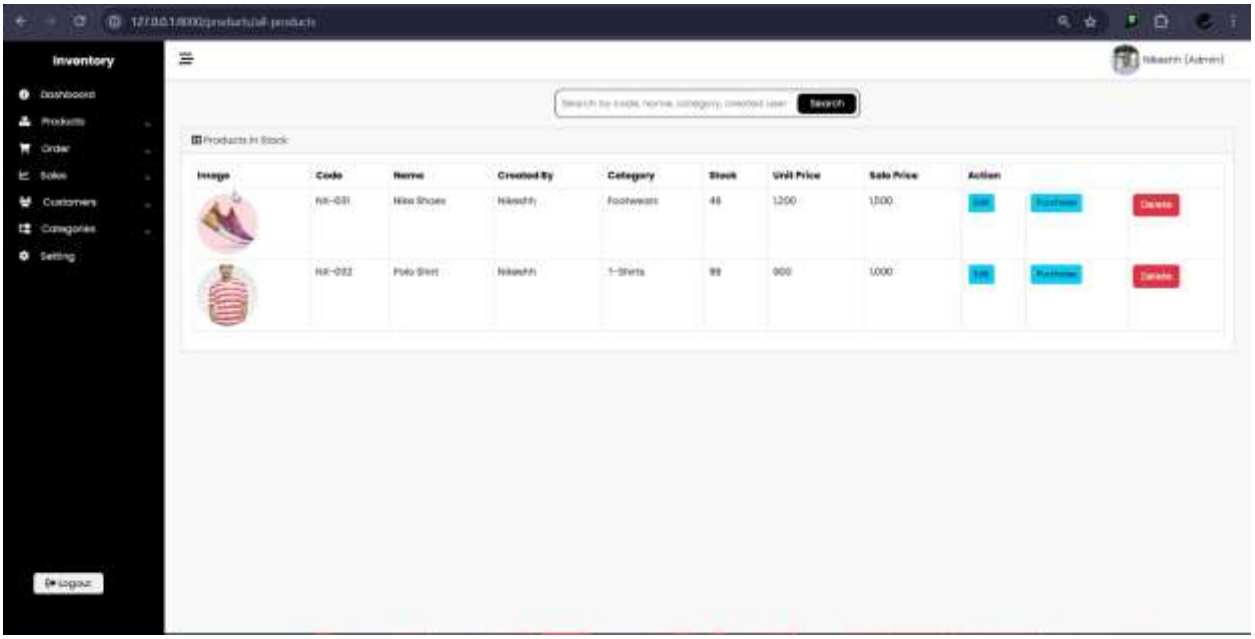
Edit Product



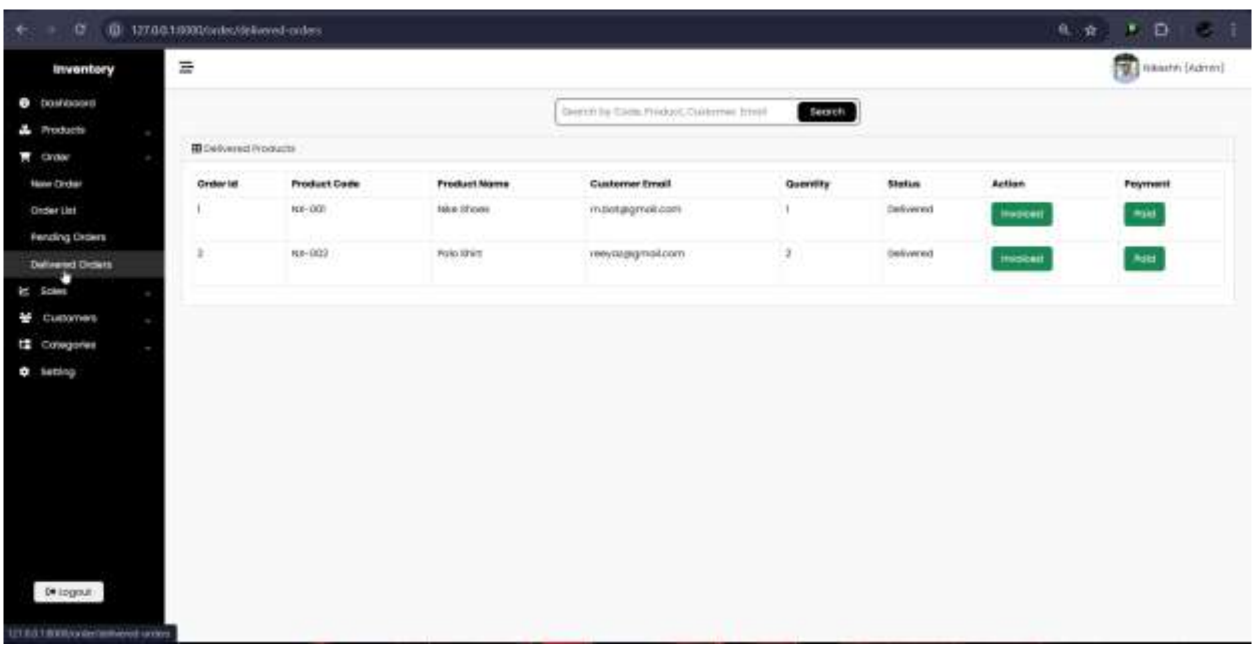
Available Product



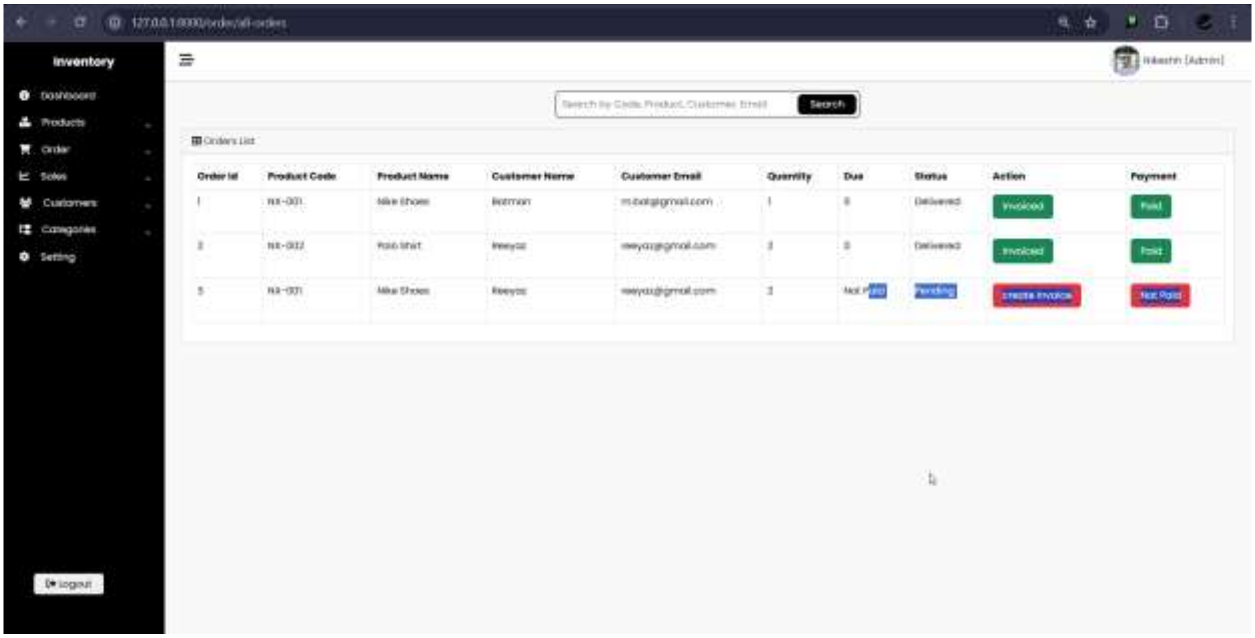
Purchase Page



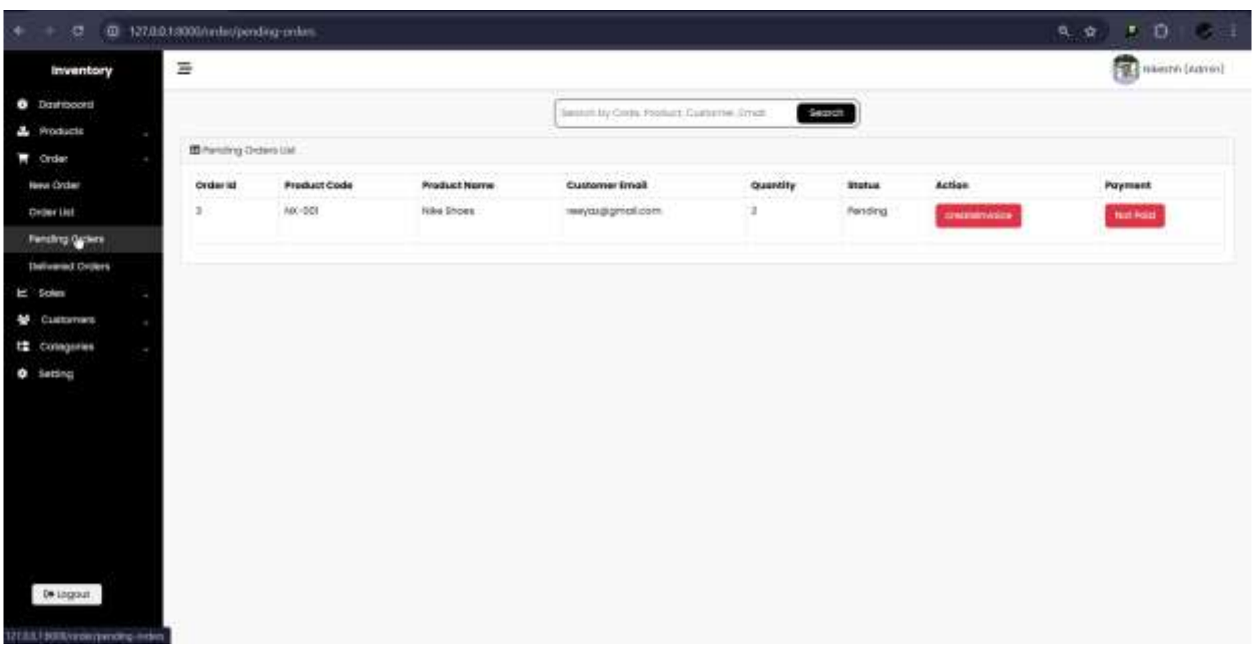
Product Stock



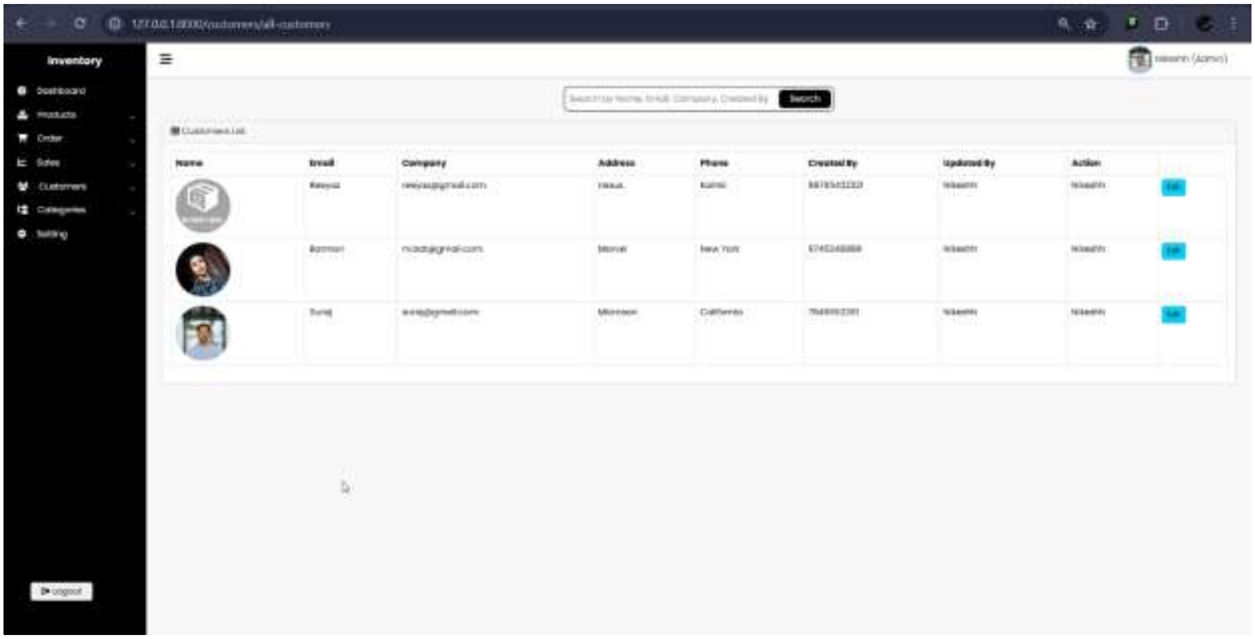
Delivered Product



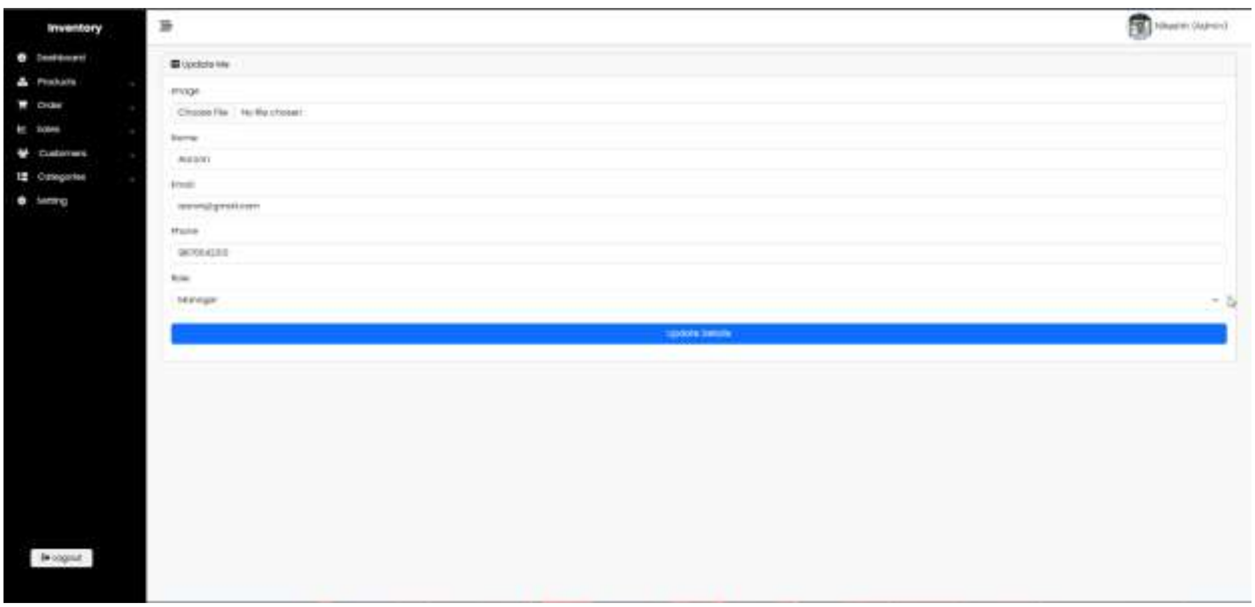
Order List



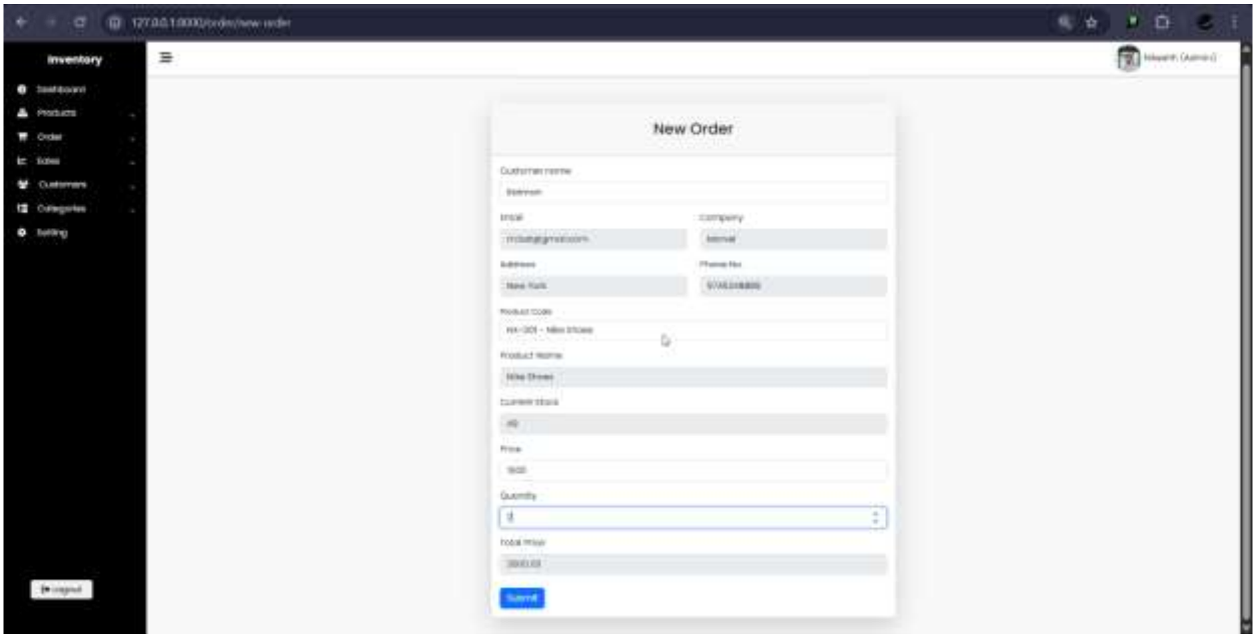
Pending Order



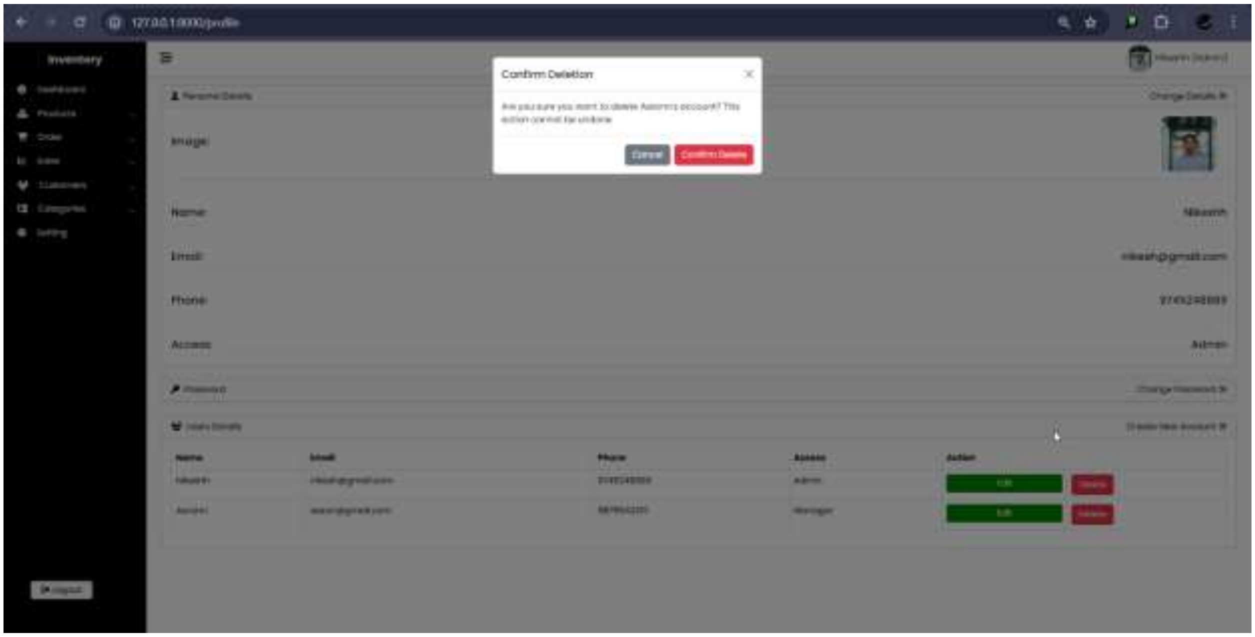
Customer List



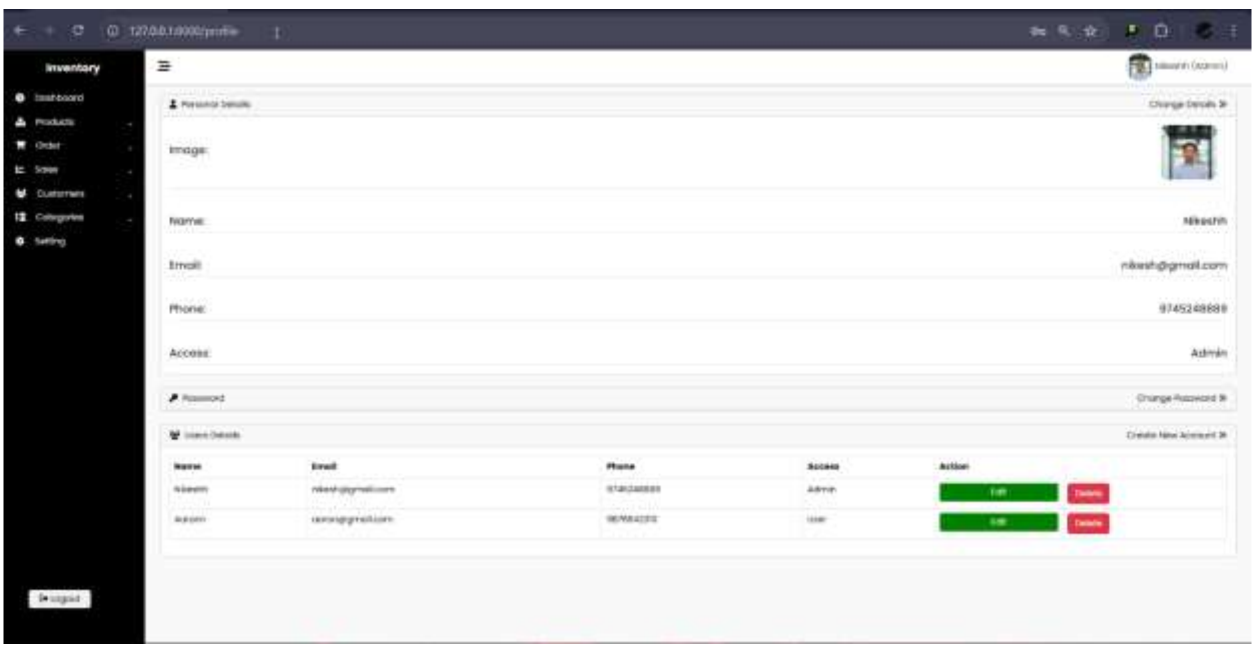
Update User



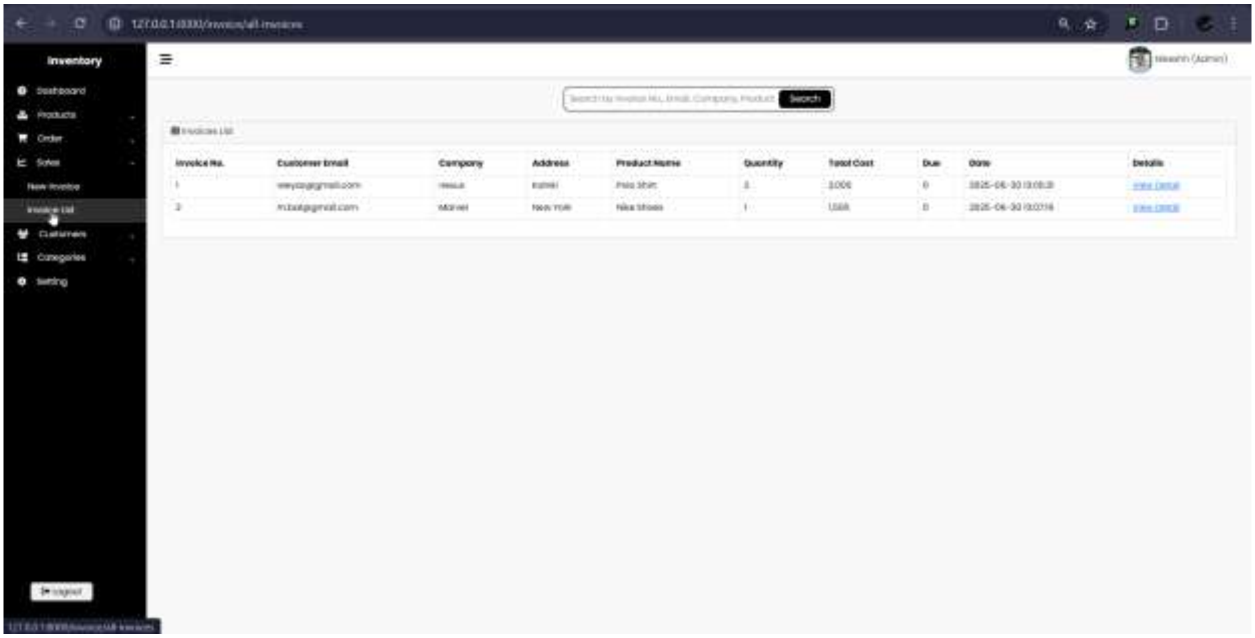
Create Order



Delete Confirmation



User Management



Invoice List